Modern Methods of Software Engineering

Homework 1

Group 37
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Question 1

Name: Login

Participating Actors: Developer, IT Manager, Financial Manager

Entry Condition: User tries to access the system.

Exit Condition: User is granted Access to the system.

Quality Condition: The system should be available and functioning.

Event Flow:

1. User tries to access the system
2. User enters his credentials
3. User presses the login button
4. User is redirected to either the main system page or the invalid credentials page

Name: Create Salary Raise Request

Participating Actors: Developer

Entry Condition: User is logged in the system

Exit Condition: A request is submitted and propagated to the IT Manager

Quality Condition: The system should be available and functioning.

Event Flow:

1. User initiates a new salary raise request
2. User fills in the required information
3. User submits the form

Name: Review Request

Participating Actors: IT Manager

Entry Condition: User is logged in the system and a request has been submitted.

Exit Condition: The request is either approved and propagated to the Financial Manager or rejected.

Quality Condition: The system should be available and functioning.

Event Flow:

1. User initiates the request review process
2. User checks the performance reports of the requester
3. User checks the salary raise priorities
4. User either approves or rejects the request

Name: Review Request

Participating Actors: Financial Manager

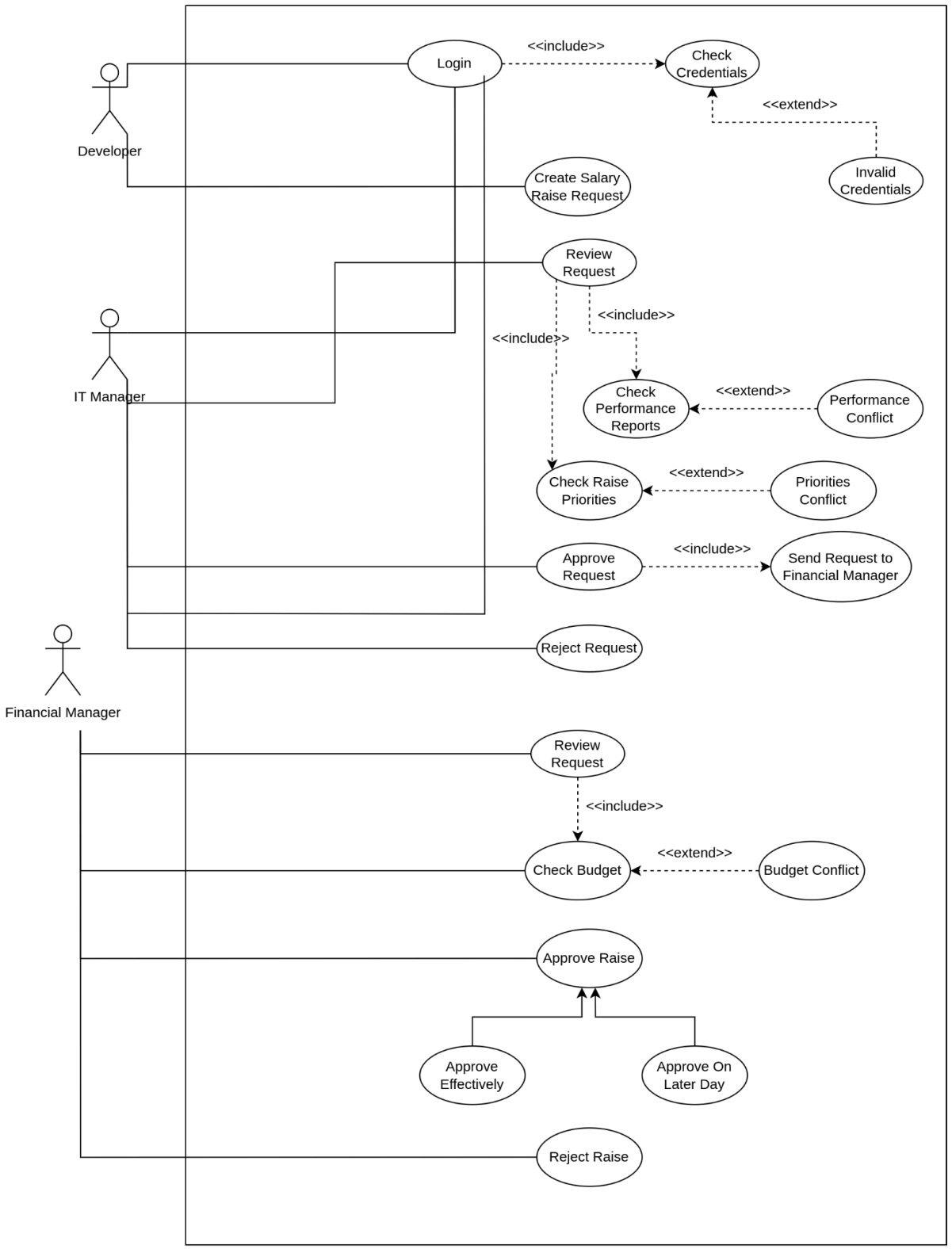
Entry Condition: User is logged in the system and a request has been approved by the IT Manager.

Exit Condition: The request is either approved (immediately or at a later date) or rejected.

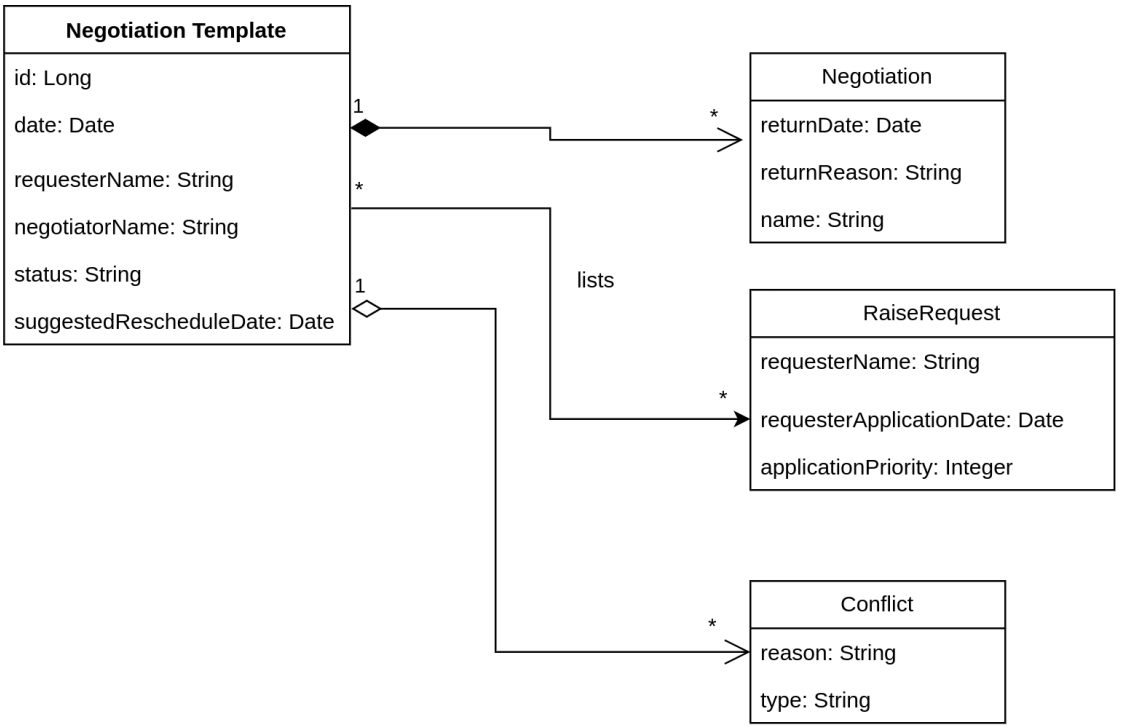
Quality Condition: The system should be available and functioning.

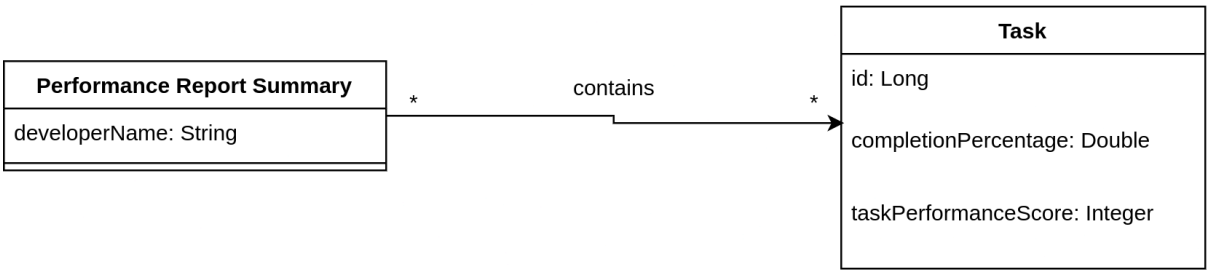
Event Flow:

1. User initiates the request review process
2. User checks the budget
3. User approves the request either immediately or at a later date, or rejects the request

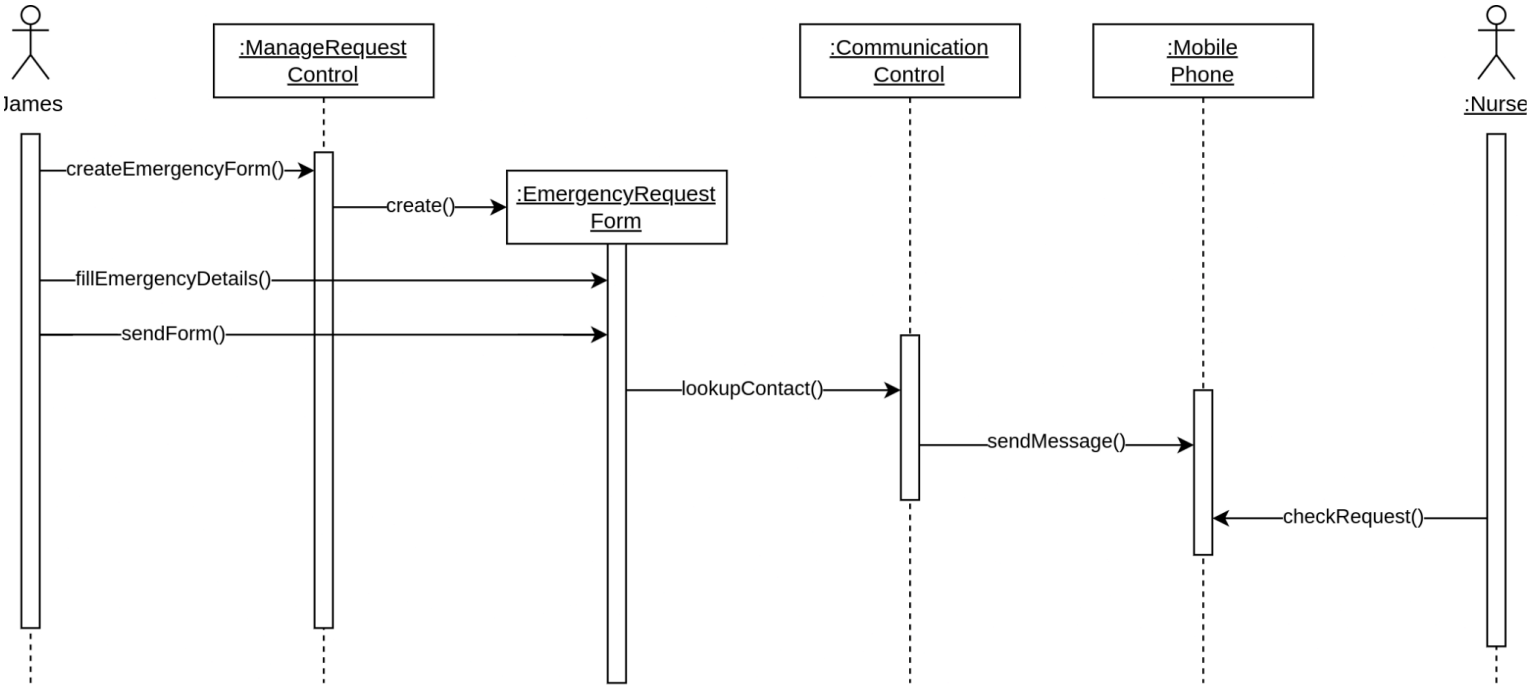

Developer
IT Manager
Financial Manager
Login
<<include>>
Check Credentials
<<extend>>
Invalid Credentials
Create Salary Raise Request
Review Request
<<include>>
<<include>>
Check Performance Reports
<<extend>>
Performance Conflict
Check Raise Priorities
<<extend>>
Priorities Conflict
Approve Request
<<include>>
Send Request to Financial Manager
Reject Request
Review Request
<<include>>
Check Budget
<<extend>>
Budget Conflict
Approve Raise
Approve Effectively
Approve On Later Day
Reject Raise

Question 2

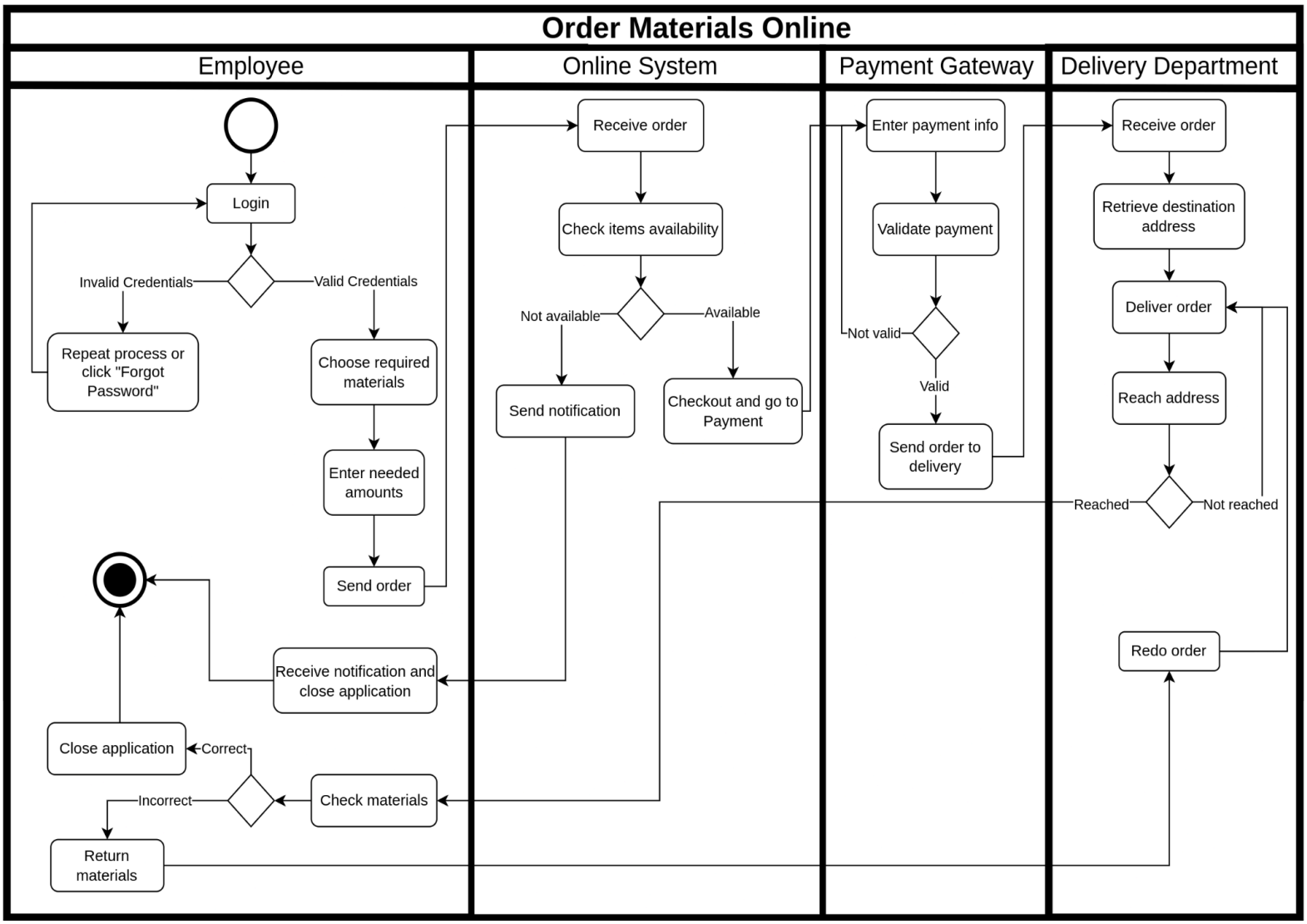

Negotiation Template
id: Long
date: Date
requesterName: String
negotiatorName: String
status: String
suggestedRescheduleDate: Date
1
*
1
*
Negotiation
returnDate: Date
returnReason: String
name: String
lists
RaiseRequest
requesterName: String
requesterApplicationDate: Date
applicationPriority: Integer
*
Conflict
reason: String
type: String
*


Performance Report Summary
developerName: String
*
contains
*
Task
id: Long
completionPercentage: Double
taskPerformanceScore: Integer

Question 3


James
:ManageRequest Control
:Communication Control
:Mobile Phone
:Nurse
createEmergencyForm()
create()
:EmergencyRequest Form
fillEmergencyDetails()
sendForm()
lookupContact()
sendMessage()
checkRequest()

Question 4


Order Materials Online
Employee
Online System
Payment Gateway
Delivery Department
Login
Invalid Credentials
Valid Credentials
Repeat process or click "Forgot Password"
Choose required materials
Enter needed amounts
Send order
Receive order
Check items availability
Not available
Available
Send notification
Checkout and go to Payment
Enter payment info
Validate payment
Not valid
Valid
Send order to delivery
Receive order
Retrieve destination address
Deliver order
Reach address
Reached
Not reached
Redo order
Receive notification and close application
Close application
Correct
Incorrect
Check materials
Return materials